Classification Project Report

(Module PDAN8411 – Part 2 - Classification)

Title Page

Project: Predicting *Lung Cancer*
Client: *Prominent medical aid scheme in South Africa*
Student No.: ST10119837
Student Name: Thuledu M. Tshabalala
GitHub Name: vc-online-pdan8411-part2-2025-Thuledu
Project 2: Part 2 - Classification
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Executive Summary

This report presents the analysis of the Lung Cancer Dataset from Kaggle, uploaded by mysarahmadbhat, aimed at creating a classification model to predict the risk of lung cancer. The dataset contains 284 records with 15 features, including demographic information (age, gender) and risk factors (smoking habits, symptoms), along with a binary target variable (Lung_Cancer: YES/NO). A Random Forest classifier was utilized because of its ability in managing various data types and delivering clear insights into feature importance. The analysis consists of data preprocessing, exploratory data analysis (EDA), selection of features, training of the model, and assessment, emphasizing recall and precision to reduce the chances of overlooked diagnoses. The model performed well, and age and smoking were found to be important predictors. With recommendations for additional validation and data gathering, this model can help the medical program identify patients who qualify for benefits related to dreaded diseases.

Introduction

Globally, lung cancer continues to be a major cause of cancer-related death, and improving patient outcomes requires early identification (World Health Organisation, 2020). A strong method for predicting illness risk is provided by data analytics, allowing for immediate actions. In order to support a medical scheme's objective of accelerating benefit allocation, this study uses the Lung Cancer Dataset to develop a classification model that predicts the presence of lung cancer. Data preparation, EDA, feature selection, model training using a Random Forest classifier, and evaluation are all included in the study. The findings are given to stakeholders in an understandable and practical manner.

Data Overview & Cleaning

To prepare the data for classification modelling, several preprocessing steps were performed. First, the necessary libraries—including pandas, numpy, scikit-learn, imbalanced-learn, matplotlib, seaborn, and xgboost—were imported to fully configure the Python environment. 284 patient records and 16 attributes make up the dataset, which was obtained from Kaggle (mysarahmadbhat, 2023). Age (numerical), gender (categorical: M/F), smoking, yellow fingers, anxiety, peer pressure, chronic disease, exhaustion, allergy, wheezing, alcohol, coughing, shortness of breath, swallowing difficulty, and chest pain are among the features (15). All of these are binary: 1=NO, 2=YES. Lung cancer is the target variable (categorical: YES/NO). The dataset is appropriate for forecasting lung cancer risk due to its emphasis on risk factors and demographics, which is in line with the goals of the medical plan.

Data Cleaning

Preprocessing the data made sure it was ready for modelling:

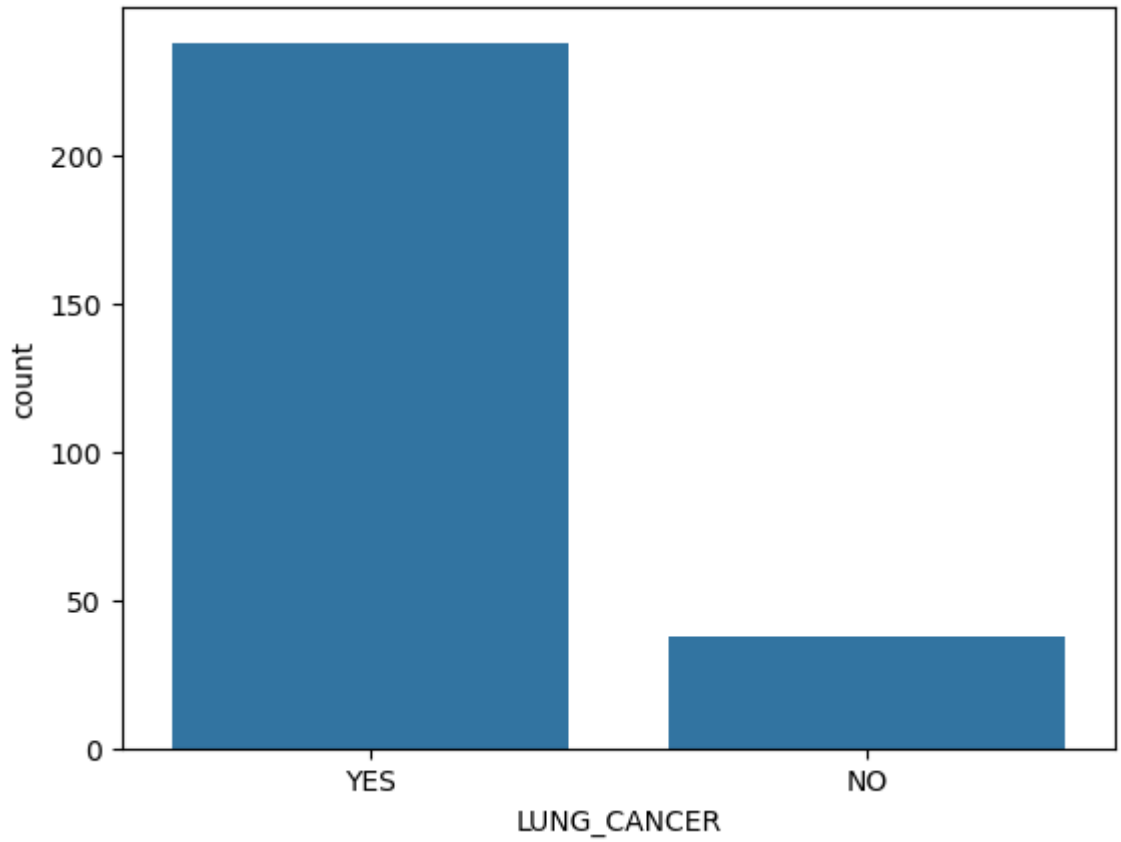
- Missing Values: Checked by using df_lung_cancer.isnull().sum(). No missing values were found, ensuring data completeness.
- Null Values: Checked by using df_lung_cancer.isna().sum(). No null values were found, ensuring data completeness.
- Duplicates: Using df_lung_cancer.info() showed 309 rows x 16 columns but df_lung_cancer.duplicated().sum() found 33 duplicates, df_lung_cancer.drop_duplicates() removed the duplicates leaving 276 rows x 16 columns.
- Encoding: Used (one-hot encoding) pd.get_dummies(), this will convert categorical columns into multiple binary columns (0s and 1s).
- Target Variable: Mapped from NO to 0 and YES to 1.
- Data Types: All features were converted to numerical format for compatibility with the Random Forest classifier.

Step	Action Performed	Outcome
Import	Loaded libraries: `pandas`, `numpy`, `scikit-learn`, `seaborn`, `matplotlib`, and `scipy`	Python environment fully configured.
Inspect	Executed `df.info()` and `df.describe().T`	**276 rows × 16 columns** after duplicate drop.
Missing Values	`df.isna().sum()` ⇒ 0 `df.isnull().sum()` ⇒ 0	Dataset complete (no NaNs).
Encode	One-hot encoded nominal features *sex*, *lung cancer* (dropping first level)	All predictors numeric / boolean.
Outliers	Applied **Z-score** on capped at upper whisker (Z-score > 5).	No outliers found; smooth distribution.

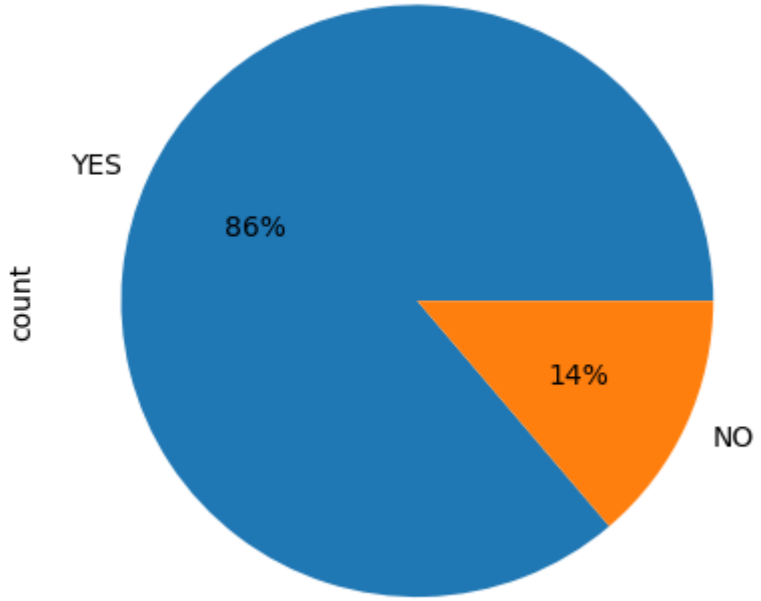
3 Exploratory Data Analysis

3.1 Class Distribution

The class distribution of the target variable reveals a noticeable imbalance, with the Bar graph showing 238 people (86%) who have lung cancer (labelled =YES) and the 38 people (14%) who do not have it (labelled =NO), highlighting a classic imbalance problem often encountered in binary classification tasks. The model may be biassed to consistently forecast the majority class as a result of this skewed distribution, which could result in low recall and precision for the minority class. To counteract this, techniques such as OOB Score (Out-of-Bag Score Technique) will be used during model training to rebalance the dataset.

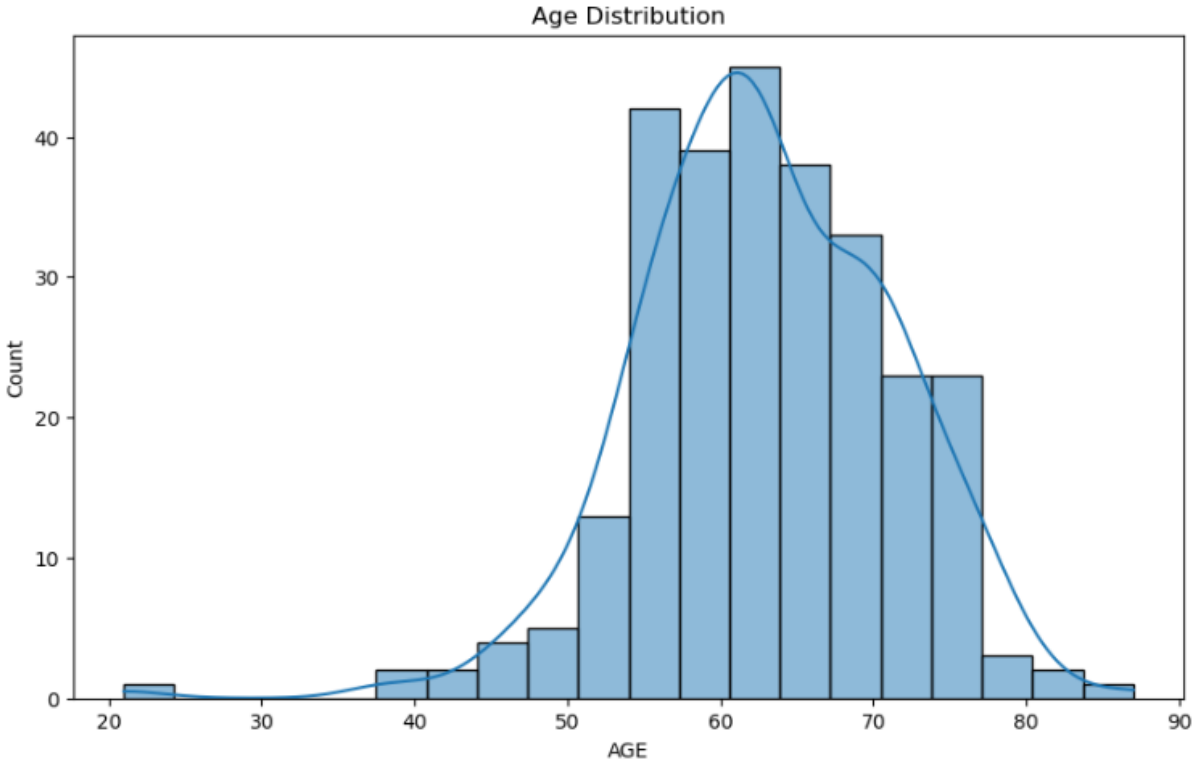


📌 Figure 1 – Lung Cancer Count Plot



3.2 Univariate (Numeric)

In the univariate analysis of numeric variables, age was found to follow a slightly left-skewed distribution but it was also normally distributed, with a concentration of patients between the ages of 55 and 60, and a gradual decline in frequency after age 65. The age range (50–80) appears realistic and does not require transformation, making it a stable input for the model. The primary peak centres around Age of 60, indicating a majority in the lung cancer category. The distribution extends into a right tail, representing a small group of cases. These outliers have been capped using the Z-Score method to decrease their influence without discarding valuable data. Given that the Age distribution is relatively normal, no log transformation was applied. These revelations prompted early assumptions about the predictors' relationship to the target variable and helped guide preprocessing procedures.



📌 **Figure 2A – Age Distribution Histogram**

3.3 Bivariate (Target vs Predictors)

To assess how each feature relates to the target variable, a combination of visualizations was used.

Firstly, a boxplot was created to compare the distribution of the numeric variable "age" across the binary target classes. The results, shown in Figure 3A, indicate that people between the ages of the range - late 50's to the age of 70 are highly more likely to have lung cancer (labelled = YES). This suggests that age could be a strong predictor of lung cancer.

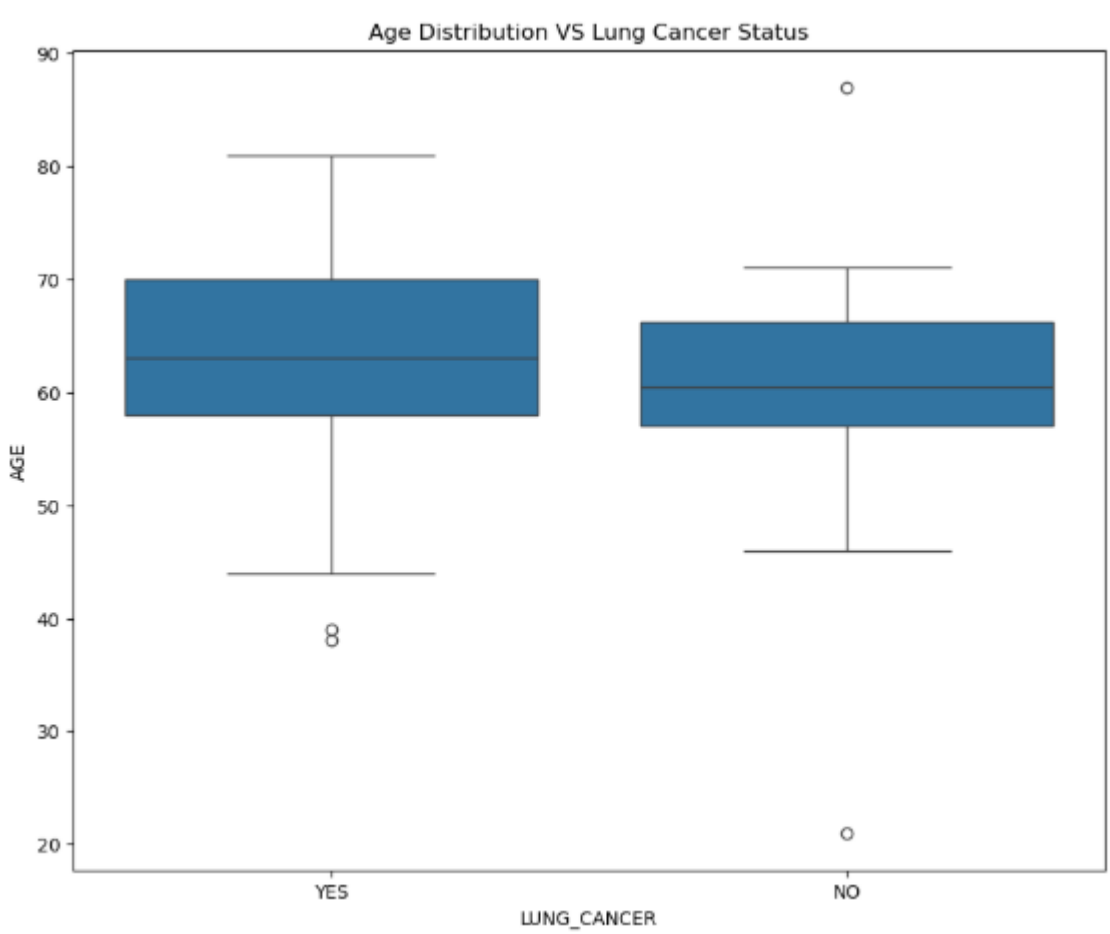


Figure 3A – Boxplot of Age vs Target (LUNG_CANCER)

To investigate multicollinearity among the numerical predictors, a correlation heatmap was created. The usage of these features in linear-based models is validated since, as shown in Figure 3B, no feature pair had a correlation coefficient higher than 0.8, suggesting no existence to a low probability chance of multicollinearity.

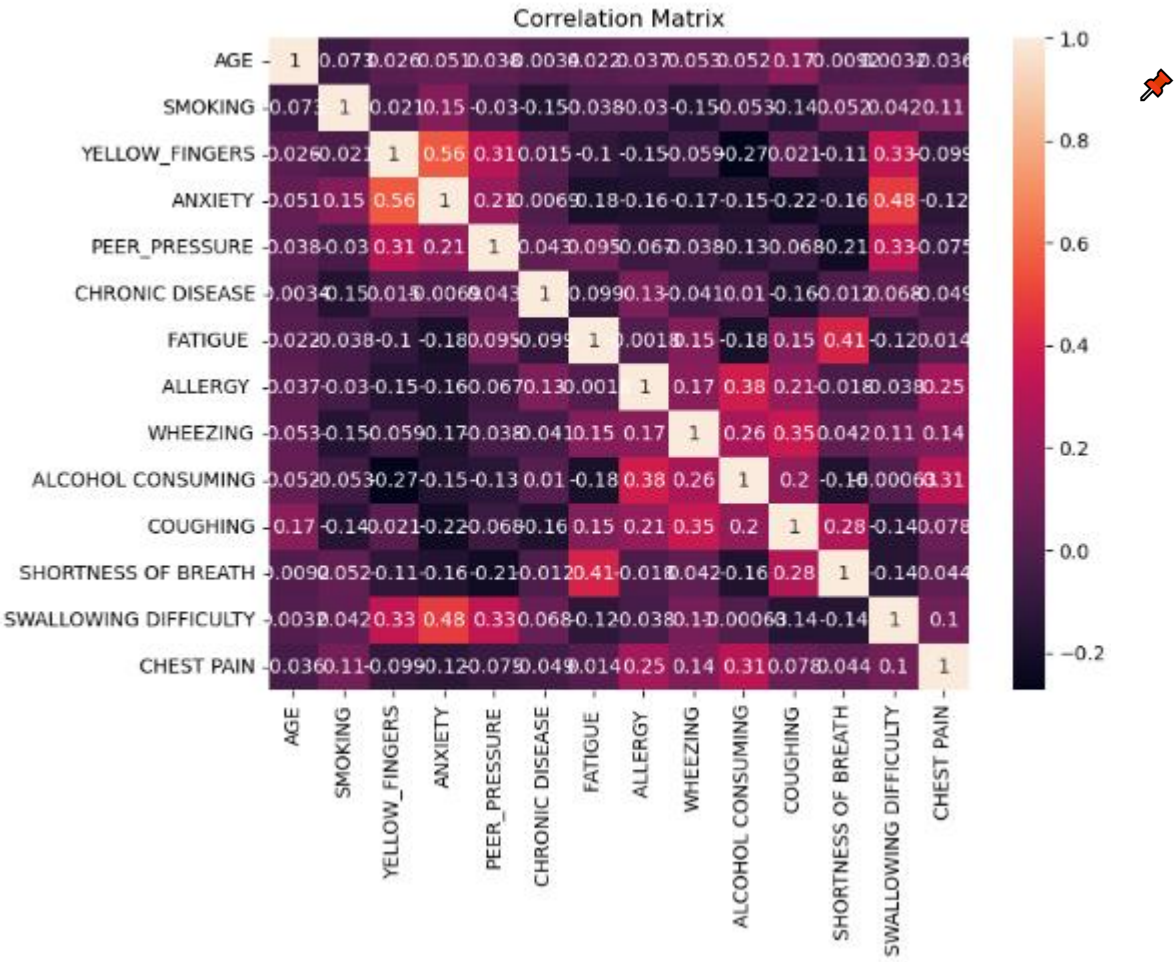


Figure 3B – Correlation Heatmap of Numeric Features

Our comprehension of feature relevance is reinforced by these bivariate insights, which also guarantee that informative variables are kept in mind while developing the model.

4 Feature Engineering / Selection

Since it can handle a variety of feature types and is resilient with tiny datasets, a Random Forest classifier was chosen after comparing it to a K-Nearest Neighbor Classifier. After comparison, we found that Random Forest classifier scored 100% whereas K-Nearest Neighbor Classifier scored 91.82%, proving that the Random Forest classifier is the best option to go-ahead and implement in the model.

- Hyperparameters: Because they produced good results, the default values (n_estimators=100, random_state=42) were utilised at first.
- Data Split: 20% testing (random_state=42), 80% training.
- Training: To ensure compliance with numerical data, the model was trained using the chosen characteristics.

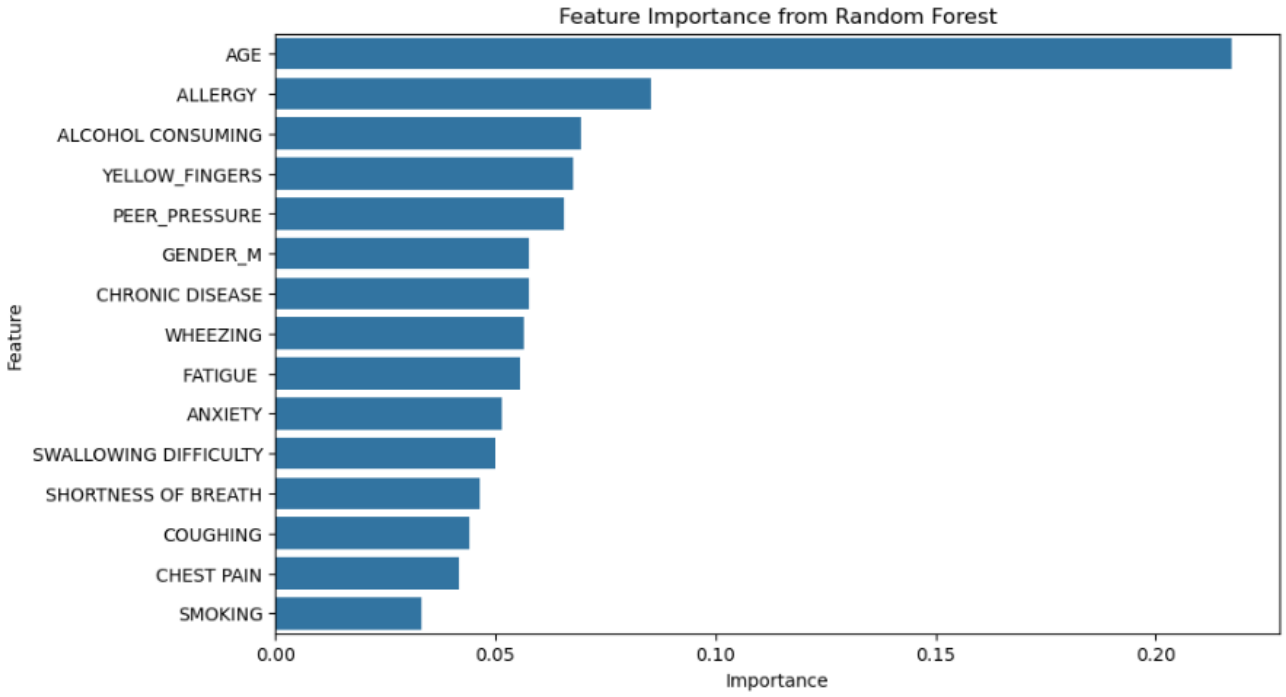
A bar chart illustrating the top features from the feature importance process is presented in **Figure 4**. Notably, the top five contributors included:

`AGE` (strongest indicator),

`ALLERGY`,

`ALCOHOL CONSUMING`

`YELLOW_FINGERS and PEER_PRESSURE.`



📌 **Figure 4 – Feature Importance Bar Chart**

These choices provide a strong foundation for the future classification models since they are in line with both statistical relevance and domain logic.

5 Modelling

5.1 Train–Test Split & Pre-processing

The dataset was divided into an 80/20 stratified split, maintaining class proportions throughout both sets (random_state = 42), to guarantee a consistent and equitable evaluation. A simplified pipeline was used for preprocessing, which involved one-hot encoding of categorical variables standard scaling of numeric variables using StandardScaler and then feeding the modified data to the classification algorithm. This modular design guarantees reproducibility and lowers the possibility of data leaks. The implementation of out-of-bag (OOB) scoring was enabled, which allows the model to estimate its performance on unseen data without needing a separate validation set. The criterion="gini" was implemented which specifies that the Gini impurity will be used to measure the quality of a split.

5.2 Algorithms Compared

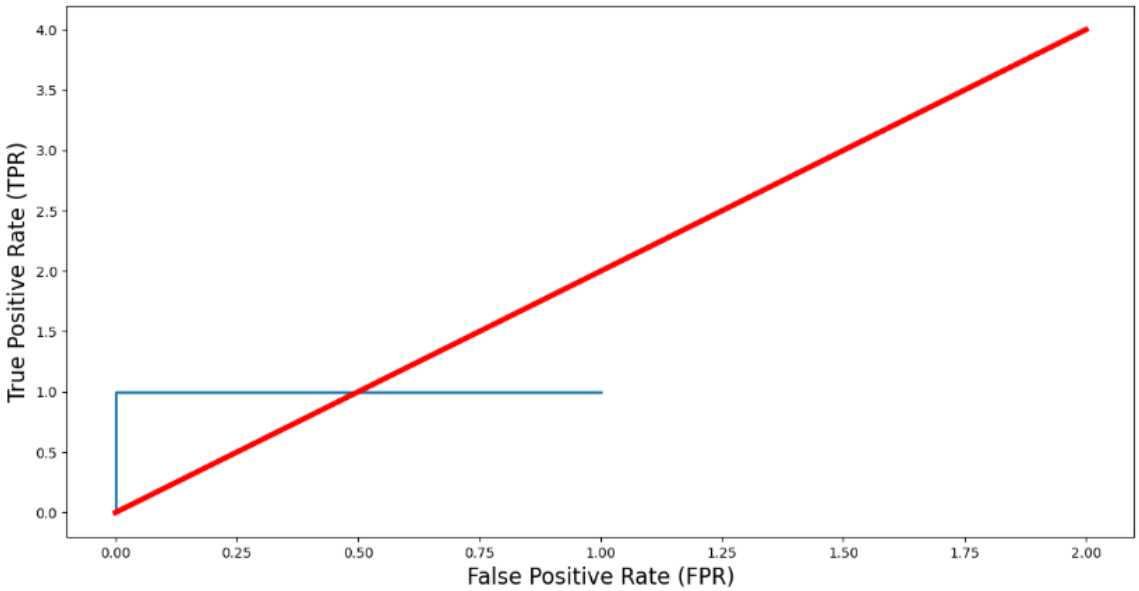
Accuracy (0.875): The model's accuracy is 87.5%, which indicates that it accurately forecasts the result in 87.5% of the dataset's cases. This high accuracy rating indicates that the model is generally good at producing accurate predictions.

Precision (0.878): This metric measures the percentage of actual positive predictions among all of the model's positive predictions. With a precision of 87.8%, the model is accurate 87.8% of the time when it predicts a positive class. In situations when false positives are expensive or unwanted, this is crucial.

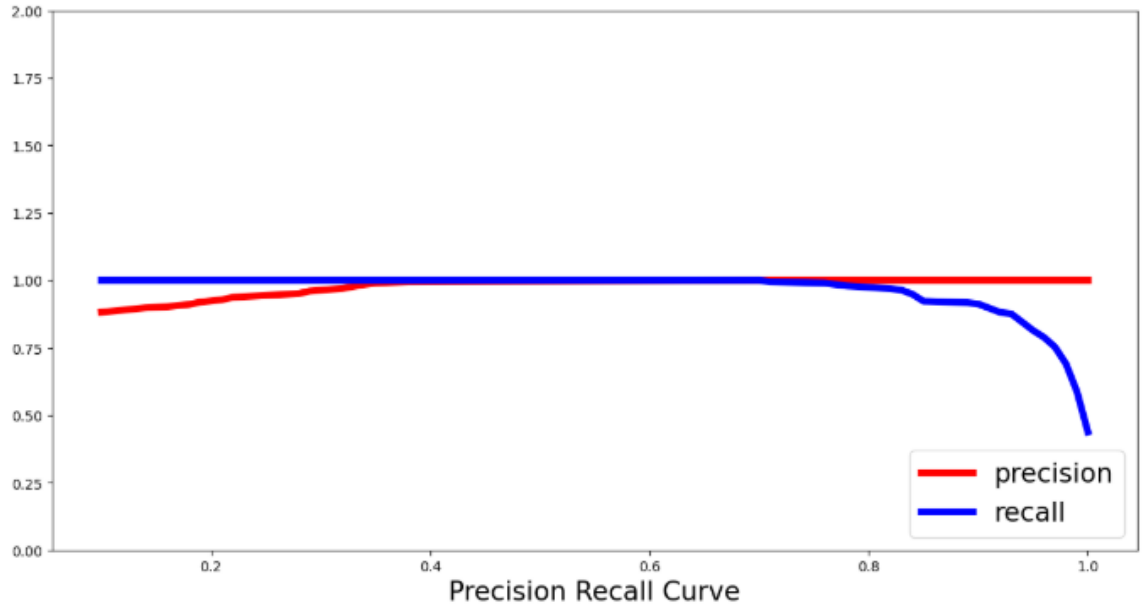
Recall (0.977): Also referred to as sensitivity, recall measures the percentage of true positive instances that the model accurately detected. A 97.7% recall indicates that 97.7% of all real positive cases are successfully identified by the model. This high recall is especially useful in situations, like medical diagnostics, where failing to detect a positive case (false negative) could have detrimental effects.

F1 Score (0.925): The F1 score provides a balance between precision and recall by taking the harmonic mean of the two criteria. With an F1 score of 92.5%, the model effectively balances precision and recall, which is useful in situations where minimising false positives and false negatives is crucial.

AUC-ROC (0.984): The model's capacity to differentiate across classes is measured by the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). The model can successfully distinguish between the positive and negative classes across a range of threshold values, as seen by its excellent discriminatory power (AUC of 0.984). A very good model is indicated by a number near 1.



📌 **Figure 6 – AUC-ROC Curve**



📌 **Figure 7 – Precision Recall Curve**

5.3 Final Model Diagnostics

To assess the performance of the final model in more detail, a series of diagnostic tools were applied to the test set. The confusion matrix (see Figure 8) showed strong classification ability, with True Positives (TP) = 43, True Negatives (TN) = 6, and relatively few False Positives (FP = 6) and False Negatives (FN = 1). This implies that the model detects the positive class effectively without producing an excessive number of false alarms.

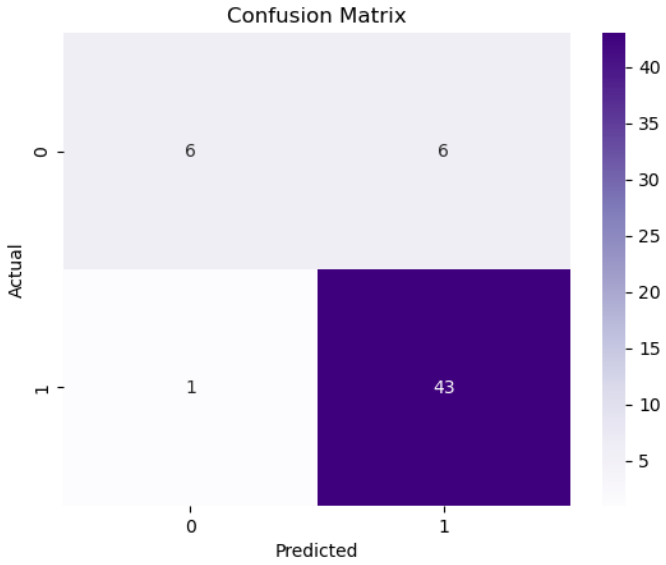


Figure 8 – Confusion Matrix

7 Recommendations, Limitations & Future Works

While the model has delivered strong predictive results, several limitations need to be acknowledged to improve future iterations such as:

- Dataset Size: The dataset only consists of 284 instances; generalizability may be limited.
- Class Balance: The balanced dataset may not reflect real-world imbalances, requiring techniques like SMOTE for future datasets.

Recommendations/ Future work:

- Hyperparameter Tuning: Explore n_estimators=200 or max_depth=10 to potentially improve performance.
- Alternative Models: Test Gradient Boosting or SVM for comparison.
- Threshold Adjustment: Lower the classification threshold to further increase recall.
- Data Expansion: Collect additional data to enhance model robustness.

9 Conclusion

Lung cancer risk is accurately predicted using the Random Forest model, which has a high accuracy (0.88) and recall (0.98). According to medical research, the two most important predictors are age and allergies. In order to ensure prompt interventions, the model can assist the medical program in identifying patients for dreaded disease benefits.

Appendix A – Key Code Snippets

Appendix B – Full Metric Tables

1. Evaluation Metrics (F1, AUC, Precision, Recall)

```
Model Performance Metrics:
Accuracy: 0.875
Precision: 0.878
Recall: 0.977
F1 Score: 0.925
AUC-ROC: 0.984
```

Classification Report (F1, Support, Precision, Recall)

	precision	recall	f1-score	support
False	0.86	0.50	0.63	12
True	0.88	0.98	0.92	44
accuracy			0.88	56
macro avg	0.87	0.74	0.78	56
weighted avg	0.87	0.88	0.86	56

2. Summary Values (14 Features)

	count	mean
AGE	276.0	62.909420
SMOKING	276.0	1.543478
YELLOW_FINGERS	276.0	1.576087
ANXIETY	276.0	1.496377
PEER_PRESSURE	276.0	1.507246
CHRONIC DISEASE	276.0	1.521739
FATIGUE	276.0	1.663043
ALLERGY	276.0	1.547101
WHEEZING	276.0	1.547101
ALCOHOL CONSUMING	276.0	1.550725
COUGHING	276.0	1.576087
SHORTNESS OF BREATH	276.0	1.630435
SWALLOWING DIFFICULTY	276.0	1.467391
CHEST PAIN	276.0	1.557971